Do Now · retrieval

Programming paradigms: applied comparison

Name: _____ Class: _____ Date: _____

Answer four retrieval questions before reviewing earlier learning.

Q1

How does OOP differ from a procedural program?

Q2

What does a local variable mean?

Q3

If a field is private-set, how can callers change it safely?

Q4

Why does `Deposit(balance,5);` not update the caller's int balance if only a return value is produced?

New Learning · key ideas

The important knowledge from today's lesson. Read, annotate, and explain it.

Core idea 1

Procedural state — The caller passes an integer balance and explicitly stores the returned value.

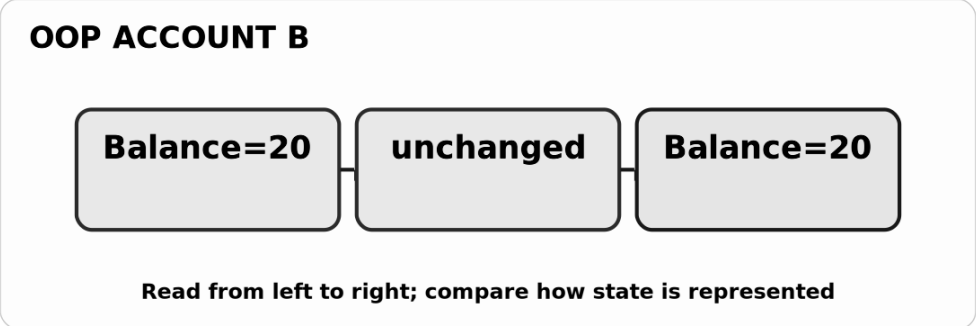
Core idea 2

Object state — Account object A stores Balance internally and Deposit changes that object's state.

Core idea 3

Separate instances — Account B stays at 20 when Account A changes to 15; the objects have independent state.

Key visual · complete concept



Annotate the key visual

Explain which labels or features show the main process.

New Learning · applying the ideas

Vocabulary · use the correct technical terms

refactoring — Restructuring code while preserving expected behaviour.

encapsulation — Controlled access to internal state.

instance — Individual object created from a class.

return value — Output returned to the caller.

Check your understanding

Explain one key process in your own words using two of the terms above.

More animated examples are available in the original online lesson.

I Do · worked example

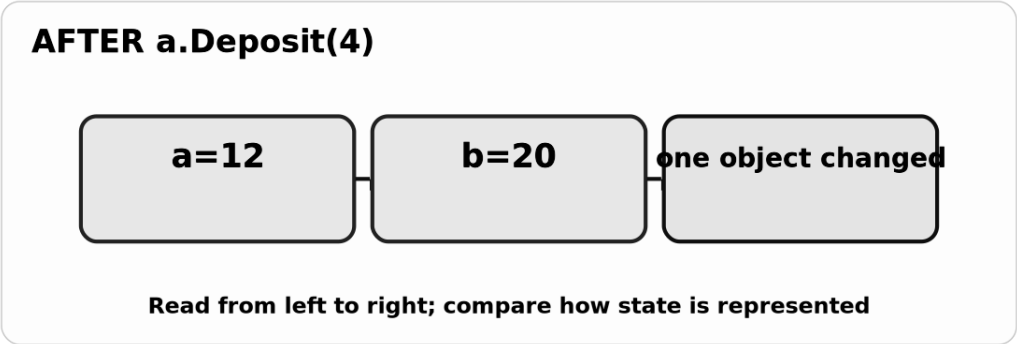
Problem

Two accounts start at 8 and 20; only first account deposits 4. Trace both.

Follow the worked method

1. Predict — Account a's Balance becomes 12.
2. Model response — Account b remains at 20 because its state is separate.

Source model · final illustrated step



Why this method works

Identify the decision, rule or invariant that makes the method correct.

We Do · Try → Discuss → Improve

TRY independently in BLACK PEN; compare methods with a partner; improve your reasoning.

Guided problem · TRY

Predict

Deposit computes and returns 10, but its result is ignored.

DISCUSS

Compare how you reached the answer. Where could someone make an error?

IMPROVE

Add one precise correction or a stronger reason. Answers are not printed here.

You Do · independent practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 1

Procedural AddFive(x) returns x+5. If caller x=2 discards the result, what is x?

Question 2

Two objects have balances A=10, B=30. A.Deposit(7) and B.Deposit(2) run. Find both balances and describe why they differ.

Check that you have used the technical terms and explained your decisions.

You Do · extended practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 3

Compare one maintenance advantage and one risk of procedural and OOP implementations for a library system.

Check that you have used the technical terms and explained your decisions.

Exit Ticket · reflect and retrieve

Exit Ticket

Calling a C# function that returns a new balance automatically changes the caller's original int variable.

Use key vocabulary from memory

1. Explain refactoring in your own words.

2. Explain encapsulation in your own words.

3. Explain instance in your own words.

Confidence and next step

How confident are you? Thumbs up / side / down

Which question would you improve after checking the classroom model?

Check answers after the lesson using the teacher-guided source models.